Fong AI: Which AI Server to Buy

A hardware recommendation for training, video generation and serving the Fong AI model. November 2026.

Summary

Recommended: a PC workstation with an NVIDIA RTX 5090 (32 GB), about $4,500-5,500. Fong AI needs to do three things, and an NVIDIA card is the best fit for all of them:

- **Training.** MiniMind's trainers and most AI training code are written for NVIDIA graphics cards (CUDA) first.
- **Video and image generation.** AnimateDiff, Wan and Stable Diffusion run several times faster on NVIDIA than on a Mac.
- **Chat for users.** Serving a 7B to 32B model is easy on any of the options in this guide.

Why upgrade from the current Mac

The current machine is a 16 GB Mac. On it:

- Video at 512x512 with 16 frames failed when memory ran out. Only 256x256 with 8 frames works.
- Pretraining from scratch stalled before step 1 because the 1.2 GB dataset filled the memory.
- Only one big model can run at a time, so the chat app, Mistral and video can't run together.
- Fine-tuning the 7B Fong AI model works, but only with 4-bit compression and memory-saving settings, so it is slow.

Recommended build: RTX 5090 workstation

Part	Choice	Why
Graphics card	**NVIDIA RTX 5090, 32 GB**	Trains 7B models with LoRA fast. Runs the Qwen 2.5 7B behind Fong AI at full precision, and 32B models at 4-bit. Makes video at 512x512 or larger.
Main memory	128 GB DDR5	Loads large training datasets easily
Processor	AMD Ryzen 9 9950X or Intel Core Ultra 9	Fast data preparation
Storage	2-4 TB NVMe SSD	Models and datasets take a lot of space
Power supply	1200 W, 80+ Gold or better	The RTX 5090 alone draws up to about 575 W
Cooling and case	Large airflow case, good CPU cooler	Steady speed during hours-long training runs
Operating system	Ubuntu 24.04 LTS	Works best with CUDA, PyTorch, Ollama and vLLM

Estimated total: about $4,500-5,500, depending on local prices.

Other options by budget

Budget	Machine	Good for	Weak point
~$2,000	Used RTX 3090 (24 GB) PC with 64 GB RAM	Cheapest real CUDA machine. Trains 7B with LoRA and runs Fong AI well.	Older and slower; video limited to medium size
~$2,000-2,500	Mini PC with AMD Ryzen AI Max+ 395 and 128 GB	Runs big models (up to 70B) quietly at low power	Little CUDA support, so training and video are hard
~$4,000	NVIDIA DGX Spark (128 GB)	Small box that runs very large models with CUDA	Slower than an RTX 5090 for 7B models and video
~$4,500-5,500	**RTX 5090 workstation (recommended)**	**Best all-round choice: training, video and chat**	Large power draw (~800 W) and heat
~$4,000+	Mac Studio (M4 Max or M3 Ultra, 128 GB or more)	Same tools used now (MLX, Ollama), silent	Training and video much slower than NVIDIA
~$10,000-12,000	RTX PRO 6000 Blackwell (96 GB) workstation	Full fine-tuning of 7B-14B models, many users at once	Expensive

What each option can run

Task	Mac 16 GB (now)	Used RTX 3090	RTX 5090	RTX PRO 6000
Chat with Fong AI 7B	Yes (4-bit)	Yes	Yes, fast	Yes, many users
Chat with a 32B model	No	4-bit, tight fit	Yes (4-bit)	Yes
LoRA fine-tune 7B	Slow, 4-bit only	Yes	Yes, fast	Yes, fast
Full fine-tune 7B	No	No	No	Yes
Video 512x512, 16 frames	No (memory)	Yes	Yes, fast	Yes, fast
Chat + video at the same time	No	Limited	Yes	Yes

Before you buy

- **Rent first to test.** A cloud RTX 5090 or A100 costs about $0.50-2 per hour (RunPod, Vast.ai). About $20 for a weekend of tests shows whether your training and video jobs run well on that card.
- **If the public will use it**, you also need a stable internet connection, a UPS for power cuts, and room for the power draw and heat (about 800 W under load).
- **Electricity.** At 800 W for 8 hours a day, the machine uses about 190 kWh a month. Check your local rate.
- **Check local prices.** These figures are mid-2026 estimates in US dollars. Graphics card prices change often, and import costs can add more.

Next steps

- Choose a budget, then get an exact parts list for that budget.
- Rent a matching cloud GPU and run the Fong AI build and a video test on it.
- Buy, install Ubuntu, NVIDIA drivers and CUDA, then move the Fong AI project across.